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AGE 418
Sensors and Controls in Precision Agriculture

Lecture 6 Compendium:
Proximal Plant and Crop Sensors

Reflectance, Fluorescence, and Machine Vision

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1 1. Why Sense the Plant?

In previous lectures, we focused on the soil. However, soil is just the environment. The **plant** is the factory.

1.1 The Integrator Concept

The crop is the ultimate sensor. It integrates all the variables (weather, pests, nutrients, water) into a single expression: its health and biomass.

- Soil tests tell us what *should* be available.
- Plant tissue tests (or sensors) tell us what the plant *actually* took up.

Real-Time Management

Proximal crop sensors allow for **Real-Time Variable Rate Application**. Instead of making a map weeks in advance, the tractor senses the crop health and adjusts the fertilizer rate instantly (milliseconds later).

2 2. Active Optical Sensors (Reflectance)

These are the most common commercial crop sensors (e.g., Trimble GreenSeeker, Ag Leader OptRx).

2.1 How they work

Unlike passive satellite sensors that rely on the sun, **Active Sensors** have their own internal light source.

1. The sensor emits rapid pulses of light in two specific bands: **Red** (absorbed by chlorophyll) and **Near Infrared** (reflected by biomass).
2. Photodiodes measure the light reflected back from the canopy.
3. The internal computer calculates vegetation indices like **NDVI** instantly.

2.2 Advantages over Satellites

- **Day/Night Operation:** Can work in total darkness because it provides its own light.
- **Cloud Independence:** Unaffected by cloud cover (crucial for Tanzanian rainy seasons).
- **Standardization:** The light source is constant, removing errors caused by sun angle or shadows.

3 3. Chlorophyll Fluorescence Sensing

Reflectance (NDVI) tells us about *biomass* and *chlorophyll content*. **Fluorescence** tells us about *photosynthetic efficiency*.

3.1 The Principle

When a plant absorbs light energy, it has three pathways: 1. **Photochemistry:** Use energy for photosynthesis (Growth). 2. **Heat Dissipation:** Release excess energy as heat. 3. **Fluorescence:** Re-emit energy as light (glow red).

3.2 Stress Detection

When a plant is under stress (e.g., herbicide damage, drought), its photosynthetic machinery shuts down. To avoid damage from excess energy, it increases **Fluorescence**.

- **Pre-Visual Detection:** Fluorescence sensors can detect stress days before the leaves turn yellow or wilt visible to the human eye.
- *Application:* Early detection of disease or herbicide resistance.

4 4. Thermal Sensing (Infrared Thermometry)

Thermal sensors measure the temperature of the canopy (T_c) relative to the air temperature (T_a).

4.1 The Physics of Cooling

Plants cool themselves through **Transpiration**. Water evaporates from the stomata, cooling the leaf (evaporative cooling).

- **Healthy/Well-watered:** Stomata are open $\rightarrow$ High transpiration $\rightarrow$ Canopy is cooler than air.
- **Stressed/Dry:** Stomata close to save water $\rightarrow$ Transpiration stops $\rightarrow$ Leaf heats up (can be hotter than air).

4.2 Crop Water Stress Index (CWSI)

Engineers use thermal data to calculate CWSI, which ranges from 0 (No Stress) to 1 (Severe Stress). This is used to trigger automated irrigation systems.

5 5. Machine Vision and AI

This is the frontier of agricultural robotics: Teaching computers to "see" like humans.

5.1 Weed Detection Systems

5.1.1 1. Green-on-Brown

Detecting green plants against brown soil.

- *Tech:* Simple color sensors or NDVI cameras.
- *Use:* Spot-spraying weeds in a fallow field (before planting).

5.1.2 2. Green-on-Green

Detecting a green weed inside a green crop (e.g., Pigweed in a Soybean field).

- *Tech:* Artificial Intelligence (Convolutional Neural Networks - CNNs).
- *Process:* The camera takes a picture. The AI analyzes leaf shape, texture, and vein patterns to classify "Crop" vs. "Weed."
- *Action:* A specific nozzle opens to spray *only* the weed.
- *Benefit:* Reduces herbicide use by up to 90%.

6 6. LiDAR (Light Detection and Ranging)

LiDAR uses laser pulses to measure distance. In agriculture, it creates a 3D Point Cloud of the crop.

6.1 Applications

1. **Biomass Estimation:** Measuring the exact height and volume of a maize or sugarcane crop to predict yield.
2. **Orchard Spraying:** Sensors measure the tree canopy shape. If there is a gap between trees, the sprayer turns off. If the tree is small, it reduces the volume.
3. **Phenotyping:** Researchers use LiDAR to measure plant architecture (leaf angle, branching) for breeding.

7 7. Leaf Wetness Sensors

Fungal diseases (like Late Blight in potatoes or Rust in beans) require free water on the leaf surface to germinate.

7.1 Electronic Leaf

A sensor designed to mimic a real leaf. It consists of a circuit board with interdigitated gold-plated traces.

- **Mechanism:** When water (dew/rain) sits on the sensor, the resistance drops.
- **Data:** It logs the "Duration of Wetness."
- **Logic:** IF (Temperature > 20řC) AND (Wetness Duration > 10 Hours) $\rightarrow$ TRIGGER FUNGICIDE ALERT.

8 8. Lecture Assessment

8.1 Part A: Simple Concept Questions

1. Explain the difference between **Active** and **Passive** optical sensors. Why are Active sensors preferred for tractor-mounted Nitrogen application?
2. Describe the working principle of the **GreenSeeker** sensor. What two bands of light does it emit?
3. Define **Chlorophyll Fluorescence**. How does it serve as an early warning system for plant stress?
4. Explain the relationship between **Stomatal Conductance** and **Canopy Temperature**. Why does a stressed plant get hot?
5. What is the **Crop Water Stress Index (CWSI)**, and how is it used in precision irrigation?
6. Differentiate between **Green-on-Brown** and **Green-on-Green** weed detection technologies.
7. How does a **LiDAR** sensor measure crop biomass?
8. What is the primary use of **Leaf Wetness Sensors** in agriculture?
9. Why is **Variable Rate Nitrogen** application based on real-time sensing often more efficient than map-based application?
10. Explain how "See & Spray" technology can reduce the environmental impact of herbicides.

8.2 Part B: Advanced Analysis & Real-World Issues

1. **Tuning a Weed Detection Algorithm:** You are building a Green-on-Green robot for Tanzanian maize fields. *Task:* Search for the open-source **"Crop/Weed Field Image Dataset"** (CWFID) or similar datasets on **Kaggle**. Explain the process of "Annotation" (Labeling). How many images of Tanzanian weeds (e.g., Striga) would you need to retrain a YOLO (You Only Look Once) model to work locally?
2. **Thermal Index Calculation:** Assume you have a thermal camera mounted on a drone over a tomato field in Arusha. *Task:* You measure a Canopy Temperature (T_c) of 32řC. The Air Temperature (T_a) is 28řC. The "Non-Water Stressed Baseline" (wet bulb) is 25řC, and the "Non-Transpiring Baseline" (dry leaf) is 35řC. Calculate the CWSI. Is this crop stressed?
$$CWSI = \frac{(T_c - T_{wet})}{(T_{dry} - T_{wet})}$$
3. **Comparing Sentinel-2 vs. GreenSeeker:** A farmer in Iringa uses a GreenSeeker handheld sensor and gets an average NDVI of 0.75 for a plot. The same day, a Sentinel-2 satellite image shows an NDVI of 0.60. *Task:* List three technical reasons for this discrepancy. (Hint: Consider atmospheric correction, pixel size vs. field of view, and active vs. passive light sources).

4. **Designing a Smart Sprayer Logic:** You are programming a microcontroller for a "See & Spray" nozzle. *Task:* Write a pseudocode logic block. *Inputs:* Camera Weed Confidence score (0-100%), Ground Speed (km/h). *Logic:* The nozzle should only open if Weed Confidence > 70%. The duration of the spray pulse must adjust based on speed to ensure the weed is hit.

5. **LiDAR for Tree Crops (Cashew/Mango):** Using **Google Earth**, look at a Cashew nut grove in Mtwara. *Task:* The trees are irregularly spaced. Explain how a tractor-mounted LiDAR system could save pesticides compared to a traditional "Air Blast" sprayer that runs continuously. Calculate the potential savings if 30% of the row length is empty space.

6. **Edge Computing for Robotics:** Explain why a weeding robot in a remote village in Morogoro must use **Edge Computing** rather than Cloud Computing for its vision system. Consider latency and internet availability.

7. **Calibration of Fluorescence Sensors:** Fluorescence signals are very weak compared to reflectance. *Task:* Describe the calibration procedure for a fluorescence sensor. Why must measurements often be taken at specific times of the day (or night) to get accurate stress readings?

8. **Economic Feasibility of Spot Spraying:** A herbicide costs TZS 50,000 per liter. A blanket application uses 4 liters/hectare. A "See & Spray" system reduces usage by 80% but costs TZS 10,000,000 to install. *Task:* Calculate how many hectares must be sprayed to break even on the investment.

9. **Integration with Irrigation:** Design a logic flow where a **CWSI** value from a thermal camera automatically triggers a drip irrigation zone valve. Include a safety check for "Is it raining?"

10. **Sensor Fusion:** Why might a single sensor (e.g., NDVI) be insufficient to diagnose a problem? Explain how combining **Thermal** (Temperature) and **Multispectral** (Color) data can help distinguish between Drought Stress and Disease Stress (which might both turn leaves yellow/wilted).